Shlok Thakkar

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Education

University of Illinois Urbana-Champaign Expected May 2028
B.S. in Computer Science + Economics, Minor in Statistics, GPA: 3.91/4.00 *Champaign, IL*
Awards: #5 Product of the Day, Product Hunt | 2nd Place, HackIllinois 2026 | 2nd Place, HERE Chicago Hackathon 2026 | Finalist, Builders Cup by Ramp 2026

Experience

Johnson Controls AI Lab Sep. 2026 – Present
Software Engineering Intern *Champaign, IL*
- Improved answer accuracy from 68% to 79% for a LangGraph multi-agent RAG chatbot over building-automation data by prototyping reranking and hybrid search (PostgreSQL full-text search + pgvector)
- Engineered a C#/.NET Model Context Protocol (MCP) server over a 241K-triple building graph, reducing SPARQL join queries from 3+ minutes to about 250 ms with an indexed triple store
- Built a 75-question retrieval eval harness and per-request tracing in PostgreSQL, so every retrieval change ships with a measured result; added 60+ unit tests (pytest, xUnit)

Stealth Startup (Seed-Stage Consumer AI) Jun. 2026 – Aug. 2026
Software Engineering Intern *New York, NY*
- Launched an agentic flight-compensation claims product from zero as 1 of 3 engineers, reaching #5 Product of the Day on Product Hunt, 350+ users, and about $20K recovered for users
- Designed and operated the full AWS backend (Lambda, Aurora PostgreSQL, Terraform, Inngest), holding a 99.9%+ success rate on real user traffic at under $0.05 per conversation
- Built the React consumer site, operator dashboard, and PostHog analytics that tracked cost per qualified claim and blended CAC across Reddit and Google Ads
- Reduced LLM research-pipeline latency by 75% (2–3 min to 30–40 s) by parallelizing 6 web-search calls with ThreadPoolExecutor
- Developed a prompt-injection defense layer that blocked 112 of 113 adversarial payloads before they reached the claims agent

Parallel Programming Laboratory, UIUC Apr. 2026 – Present
Undergraduate Research Assistant *Champaign, IL*
- Engineered custom operators and broadcasting in C++ for a Charm++ distributed NumPy array, and redesigned 3 core runtime APIs for Reconverse, a ground-up rewrite of the Charm++ parallel runtime

Disruption Lab @ UIUC Aug. 2025 – Present
Tech Lead (previously Software Developer) *Champaign, IL*
- Led 2 twelve-week AI course-intelligence projects for university clients, directing engineers on a RAG pipeline with Pinecone embeddings and a Neo4j concept-prerequisite knowledge graph
- Reduced graph query latency by 60% (300 ms to 120 ms) for a 40+ user platform by restructuring the Neo4j data model and Cypher queries behind 15+ Flask REST endpoints
- Expanded tax-schema coverage from 7 to 27 states for a Fortune 500 financial-services client by building a Python/lxml XSD parser, validated by 43 unit tests

Projects

greenroom | *Go, MCP, macOS VMs, LLM Evals* greenroom.shlokthakkar.com
- Created a service that gives AI coding agents their own ephemeral macOS VM to build a branch, run the app, screenshot it, and verify the fix before it opens a PR; 50+ person waitlist
- Minimized the verifier's false-pass rate from 15% to 0% on a planted-fault benchmark by adding an evidence contract, while cutting p95 verdict time from 9.5 to 3.4 minutes; 1,600+ Go tests

Switchboard | *TypeScript, Cloudflare Workers, Durable Objects, MCP* GitHub
- Built a shared task channel where Claude Code and Codex sessions across machines claim GitHub Issues as tasks, with a TypeScript CLI, MCP tools, and one SQLite-backed Durable Object per repo on a Cloudflare Worker
- Developed a streaming proxy over the Anthropic, OpenAI, and Gemini APIs that masks secrets and records every turn at under 2 ms p99 overhead (7,200 of 7,200 turns captured, no leaks); 298 tests across CLI and Worker

Technical Skills

Languages: TypeScript, Python, Go, C++, C#, SQL, Java
AI & LLM Systems: LLM Agents, RAG, LangGraph, MCP, Hybrid Search, pgvector, Pinecone, LLM Evals, Prompt-Injection Defense
Backend & Distributed Systems: Distributed Systems, Parallel Computing, REST APIs, PostgreSQL, Neo4j, Cloudflare Workers, Flask, .NET
Cloud & Tools: AWS (Lambda, Aurora), Terraform, Inngest, React, PostHog, Git, pytest, xUnit
Relevant Coursework: Algorithms & Models of Computation, Computer Architecture, System Programming, GPU Programming (CUDA)